Yifan Zhu

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Research Profile

Computer Science Ph.D. candidate developing **LLM alignment and agent systems** that reason about belief, perspective, uncertainty, and shared understanding. My research combines **post-training**, **Theory of Mind and epistemic reasoning**, and **multimodal perception**, with experience leading projects from formalization and dataset construction through model training, evaluation, and real-time research prototypes. Seeking a **Summer 2027 Research Scientist or Research Engineer internship** focused on LLMs, agents, or alignment.

Education

Brandeis University *Expected Spring 2028*
Ph.D. in Computer Science

- **Advisor:** Prof. James Pustejovsky
- **Research:** LLM alignment, Theory of Mind and epistemic reasoning, multimodal agent systems

Georgetown University *2019 – 2021*
M.S. in Computational Linguistics

University College London *2015 – 2016*
M.A. in Linguistics

Xiamen University *2011 – 2015*
B.A. in Philosophy, Chinese Language and Literature

Research Experience

Project: Friction-Based Alignment & Theory of Mind *Oct 2025 – Present*

- Developed and evaluated **friction-aware post-training objectives** against DPO, conditioning language-model interventions on instance-level epistemic risk rather than preference labels alone; across synthetic Minecraft interactions and human HCRC MapTask dialogue, and using 4-bit QLoRA with **Qwen3.5-27B**, FAR and FTR increased MapTask intervention F1 from **0.344 to 0.416–0.417**, reduced warranted-context ECE from **0.432 to 0.178–0.227**, and remained substantially more stable across three runs, with F1 standard deviations of **0.023 and 0.009** versus **0.106** for DPO. (**EMNLP 2026**)
- Developed a four-identifier **Theory-of-Mind representation** of first- and second-order referential beliefs, making silent epistemic divergence directly computable; incorporating the second-order channel increased held-out misunderstanding-detection recall from **26.1% to 65.2%**, with human agreement of κ = **0.80/0.77** on inferred second-order beliefs. (**EMNLP 2026**)
- Extended Frictive Policy Optimization to **perceptually asymmetric dialogue**; across GPT-5, Qwen3.6-35B, and Llama-4-Scout, participant-specific reasoning achieved **0.903–0.955 F1**, compared with **0.247–0.434** under omniscient access (**SIGDIAL 2026**).
- Built an end-to-end, multi-GPU research pipeline covering GPT-5 belief-state extraction, deterministic friction computation, QLoRA post-training, vLLM inference, calibration and Brier analysis, bootstrap confidence intervals, multi-seed evaluation, and order-debiased LLM judging. (**EMNLP 2026**)

Selected publications: Mind the Gap: Theory-of-Mind-Grounded Friction for Epistemic Alignment (EMNLP 2026, first author); *From Propositional to Perceptual Asymmetry: Extending FPO to Asymmetric Partial Information Dialogue* (SIGDIAL 2026, first author).

Project: Common Ground Tracking & Epistemic Modeling *Mar 2024 – Present*

- Introduced the **Distributed Partial Information Puzzle (DPIP)**, a collaborative benchmark for evaluating LLM and symbolic belief tracking under disjoint information; released a multimodal dataset of **40 participants across 10 groups** with aligned annotations of speech, gesture, object interaction, and action.
- Developed an executable **evidence-based dynamic epistemic logic** framework that converts multimodal observations and dialogue events into updates over participant beliefs, intentions, and common ground.
- Implemented the block-description transfer component of **TRACE**, a real-time multimodal common-ground tracking system demonstrated live at **NAACL 2025**, integrating multiparty audio and video evidence for downstream symbolic reasoning.
- Led dataset design, annotation, and adjudication for common-ground and friction corpora, including annotation-schema development, annotator training, inter-annotator validation, and quality-control workflows.

Selected publications: Distributed Partial Information Puzzles (LREC 2026, first author); *Multimodal Common Ground Annotation for Partial Information Collaborative Problem Solving* (ACL-ISO ISA Workshop 2025, first author); *TRACE: Real-Time Common Ground Tracking* (NAACL 2025 Demo, co-author); *Common Ground Tracking in Multimodal Dialogue* (LREC-COLING 2024, co-author).

Project: Multimodal Perception & Situational Awareness *May 2023 – Present*

- Built live and offline **2D and 3D multimodal perception pipelines** integrating speech recognition, speaker diarization, object detection, pose estimation, gaze, gesture, posture, facial behavior, co-attention, and action recognition, converting synchronized audio, video, and object-interaction streams into structured events for downstream reasoning.
- Developed multimodal situational-awareness models for collaborative problem solving using late-fusion integration of verbal and nonverbal features, including classifiers for engagement, joint attention, and collaborative behavior.
- Integrated perception and reasoning components into **real-time research prototypes and live demonstrations** for classroom collaboration and human–AI teaming.

Selected publications: Modeling Theory of Mind in Multimodal HCI (HCII 2024, first author); *Multimodal Situational Awareness in Collaborative Learning* (HCII 2025, first author); *Speech Is Not Enough: Interpreting Nonverbal Behavior in Multiparty Dialogue* (AAAI 2025, co-author).

Project: Computational Semantics & Cross-Lingual UMR *2022 – Present*

- Developed executable semantic representations that ground **Generative Lexicon event structures** in explicit preconditions and postconditions, supporting symbolic updates over actions, states, and participant knowledge.
- Conducted sentence- and document-level Chinese Universal Meaning Representation annotation and developed analyses of language-specific challenges involving verb compounds, aspect, and anaphora; proposed compound-annotation solutions **incorporated into UMR community guidelines**.

Selected publications: Lexical Event Models for Multimodal Dialogue (HCII 2024, co-author);*UMR Annotation of Chinese Verb Compounds* (UMR Workshop 2023, co-author).

Software & Datasets

- **DPIP Dataset:** multimodal collaborative-dialogue dataset for studying belief tracking under asymmetric partial information; includes aligned speech, gesture, object-interaction, and action annotations. Zenodo, 2026. https://doi.org/10.5281/zenodo.18626417
- **TRACE:** real-time multimodal common-ground tracking system integrating speech, visual perception, and symbolic reasoning; demonstrated at NAACL 2025. https://github.com/csu-signal/TRACE
- **FPO Training and Evaluation Pipeline:** research codebase for friction-aware post-training, including belief-state extraction, deterministic supervision generation, QLoRA and DPO training, vLLM inference, calibration analysis, bootstrap evaluation, and multi-model judging.

Professional Service & Leadership

- **Reviewer:** ACL Rolling Review (May 2026 cycle — AACL 2026, EMNLP 2026); LREC 2026; CV4Edu Workshop @ CVPR 2026.
- **Mentorship:** Trained and led annotation teams; designed inter-annotator validation and adjudication workflows for common-ground and friction datasets.
- **Human-subjects research:** Authored IRB protocol, consent, and instrument materials for a VR study of collaborative object tasks (Meta Quest).

Technical Skills

- **LLM Post-Training:** PyTorch, Hugging Face Transformers, TRL, PEFT, LoRA/QLoRA, DPO, PPO, GRPO, RLHF, reward modeling
- **LLM Systems:** vLLM, open-weight model deployment, prompt and data pipelines, LLM-based annotation
- **Multimodal AI:** speech recognition and diarization, object detection, pose estimation, gaze, gesture and posture extraction, late fusion
- **Infrastructure:** multi-GPU training, Linux
- **Programming & Tools:** Python, Git, OpenCV, Detectron2, MMPose, Whisper, pyannote.audio, LaTeX